

Memo.

Section A

- | | |
|------|------|
| 1. d | 5. e |
| 2. d | 6. b |
| 3. c | 7. f |
| 4. d | 8. e |

Section B

(9.1) $\frac{3}{\sqrt{1-9x^2}}$

(9.2) $\frac{1}{t} + 3^t \ln 3 + \sec^2 t$

(9.3) $8t/4t^2 = 2/t$

(9.4) $e^{\cos x^2} \times (-\sin x^2) \times (2x)$

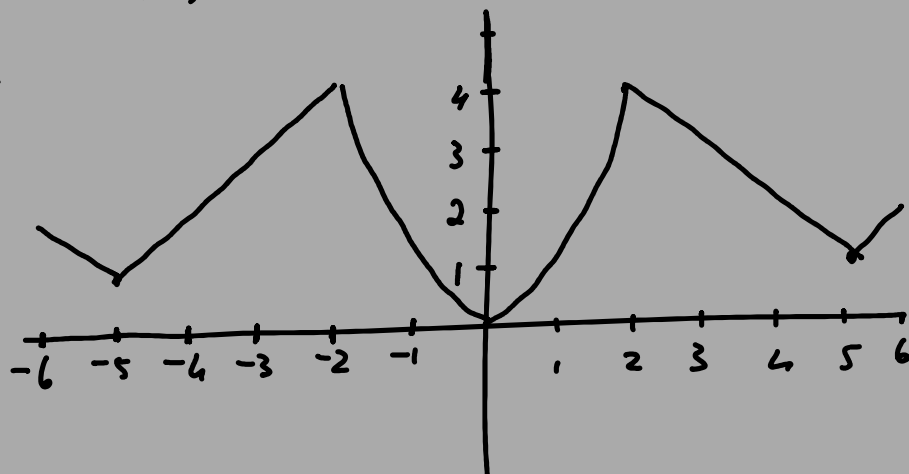
(9.5) $e^{2\cos x} \times (-2\sin x) = 2 e^{\cos x} \times e^{\cos x} \times (-\sin x)$

10. $g(4)=3$ and $g'(4)=1/4$

11. 8

12. $(-1, 2)$ and $(5, 6)$

13.



14 $a=0$

15 $x=0$ and $x=-2$

16 (16.1) see p.280

(16.2) see p.280.

Section C

Q17 (a) Form of limit: $(\infty - \infty)$

Multiply by $\frac{\sqrt{4x^2 - 3x} - 2x}{\sqrt{4x^2 - 3x} - 2x}$ and obtain

$$\lim_{x \rightarrow -\infty} \frac{-3x}{\sqrt{4x^2 - 3x} - 2x} \quad \left(\frac{\infty}{\infty} \right)$$

$$= \lim_{x \rightarrow -\infty} \frac{-3x}{\sqrt{4x^2(1 - 3/4x)} - 2x}$$

$$= \lim_{x \rightarrow -\infty} \frac{-3x}{-2x \sqrt{1 - 3/4x} - 2x}$$

$$\sqrt{4x^2} = |2x| = -2x, \\ \text{Since } x < 0$$

Factor out $-2x$, to obtain

$$\frac{3}{2} \lim_{x \rightarrow -\infty} \frac{1}{\sqrt{1 - 3/4x} + 1} = \frac{3}{2} \left(\frac{1}{1+1} \right) = \frac{3}{4}.$$

$$(b) \lim_{x \rightarrow 0} \frac{\sin 3x}{5x^2 - 4x} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \frac{3}{5x - 4}$$

$$= \lim_{x \rightarrow 0} \frac{\sin 3x}{3x} \times \lim_{x \rightarrow 0} \frac{3}{5x - 4}$$

$$= 1 \times \left(-\frac{3}{4} \right)$$

$$= -\frac{3}{4}.$$

May also use L'Hospital's rule, but you must indicate if you use it.

Question 18: $-1 \leq \sin \frac{\pi}{x} \leq 1$ for $x > 0$

$$\Rightarrow e^{-1} \leq e^{\sin \frac{\pi}{x}} \leq e^1 \text{ for } x > 0$$

$$\Rightarrow \frac{\sqrt{x}}{e} \leq \sqrt{x} e^{\sin \frac{\pi}{x}} \leq \sqrt{x} e \text{ for } x > 0$$

Since $\lim_{x \rightarrow 0^+} \frac{\sqrt{x}}{e} = 0 = \lim_{x \rightarrow 0^+} \sqrt{x} e$, it follows from the Squeeze Theorem that

$$\lim_{x \rightarrow 0^+} \sqrt{x} \sin \frac{\pi}{x} = 0$$

Question 19: See Example 8, p. 286.

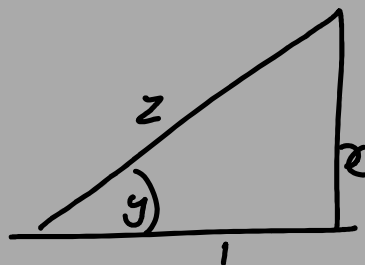
Question 20: $\tan^{-1} x = y \Leftrightarrow \tan y = x$ and $y \in (-\frac{\pi}{2}, \frac{\pi}{2})$

$$\Rightarrow \frac{d}{dx} \tan y = \frac{d}{dx} x$$

$$\Rightarrow \sec^2 y \frac{dy}{dx} = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\sec^2 y} = \cos^2 y$$

$$\tan y = x = \frac{x}{1}$$



$$\Rightarrow \text{by Pythagoras: } z^2 = 1 + x^2 \Rightarrow z = \sqrt{1 + x^2}$$

$$\text{and } \cos y = \frac{1}{\sqrt{1 + x^2}} \Rightarrow \cos^2 y = \frac{1}{1 + x^2}$$

$$\text{and } \frac{dy}{dx} = \cos^2 y = \frac{1}{1 + x^2}.$$

$$(b) \frac{dy}{dx} = \frac{-\sin x}{1 + \cos^2 x}$$

Question 21

Theorem 4, p. 157.